Name:

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ID H Vector D Vector

H Vector (A/m)

0	$\vec{H} = \vec{a}_x - (x+y)\,\vec{a}_y$	5	$\vec{H} = \vec{a}_x + y\,\vec{a}_z$
1	$\vec{H} = xy\,\vec{a}_x + \vec{a}_z$	6	$\vec{H} = x^2\,\vec{a}_y - \vec{a}_z$
2	$\vec{H} = y\,\vec{a}_y + 2\,\vec{a}_z$	7	$\vec{H} = y\,\vec{a}_x - \vec{a}_y + \vec{a}_z$
3	$\vec{H} = (x+z)\,\vec{a}_x + 2\,\vec{a}_y$	8	$\vec{H} = (x+y)\,\vec{a}_x + 2x\,\vec{a}_z$
4	$\vec{H} = \vec{a}_y + x^2\,\vec{a}_z$	9	$\vec{H} = 2x^2\,\vec{a}_x + z\,\vec{a}_y$

D Vector (C/m^2)

0	$\vec{D}_1 = \vec{a}_r - 4\,\vec{a}_\theta + \vec{a}_\varphi$	5	$\vec{D}_1 = 3\,\vec{a}_r - \vec{a}_\theta + 2\,\vec{a}_\varphi$
1	$\vec{D}_1 = \vec{a}_r - \vec{a}_\theta$	6	$\vec{D}_1 = 6\,\vec{a}_\theta + 2\,\vec{a}_\varphi$
2	$\vec{D}_1 = \vec{a}_r + \vec{a}_\theta + 5\,\vec{a}_\varphi$	7	$\vec{D}_1 = -2\,\vec{a}_r - \vec{a}_\theta + \vec{a}_\varphi$
3	$\vec{D}_1 = 2\,\vec{a}_r + 3\,\vec{a}_\varphi$	8	$\vec{D}_1 = \vec{a}_r - 4\,\vec{a}_\theta - \vec{a}_\varphi$
4	$\vec{D}_1 = 2\,\vec{a}_r - \vec{a}_\varphi$	9	$\vec{D}_1 = \vec{a}_r - 5\,\vec{a}_\varphi$

Questions

1) Use the given $\vec{H}$ and Ampere's Law $\left(\oint \vec{H}\cdot\vec{dl} = I_{enclosed}\right)$ to find the total current enclosed within the contour given below. Coordinate points are in meters.

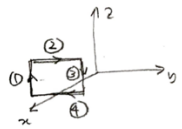
Contour: $(2,-1,0) \rightarrow (2,-1,2) \rightarrow (2,1,2) \rightarrow (2,1,0) \rightarrow (2,-1,0)$

$\vec{H} = (x+y)\,\vec{a}_x + 2x\,\vec{a}_z$

For the given contour $\vec{dl} = dy\,\vec{a}_y + dz\,\vec{a}_z$

$\vec{H}\cdot\vec{dl} = 2x\,dz$

$I_{enc} = \int_1 + \int_2 + \int_3 + \int_4$



$= \int_0^2 2x\,dz + 0 + \int_2^0 2x\,dz + 0$

$= 0$

2) Consider two regions in space separated by the surface r=5 (m). Region 1 ($r \leq 5$) is characterized by ε_r=2.5. The rest of space (region 2) is air. Given the electric flux density in region 1 as $\vec{D}_1$ (from the table), find the complete vector expression of the electric field $\vec{E}_2$ at the interface. No surface charges exist on the boundary.

Apply boundary conditions $E_{1t} = E_{2t}$, $D_{1n} = D_{2n}$

$\vec{D}_1 = 2\,\vec{a}_r + 3\,\vec{a}_\varphi$ $\quad \vec{a}_n = \vec{a}_r \Rightarrow \vec{D}_{1n} = 2\,\vec{a}_r$, $\vec{D}_{1t} = 3\,\vec{a}_\varphi$

$\vec{D}_{2n} = \vec{D}_{1n} = 2\,\vec{a}_r$

$\vec{E}_{2n} = \frac{2}{\epsilon_0}\,\vec{a}_r$

$\vec{E}_{2t} = \vec{E}_{1t} = \frac{3}{2.5\epsilon_0}\,\vec{a}_\varphi$

$\Rightarrow \vec{E}_2 = \frac{2}{\epsilon_0}\,\vec{a}_r + \frac{3}{2.5\epsilon_0}\,\vec{a}_\varphi$